Week #8 : 19th to 23rd OCtober

الأسبوع #: من - إلى

Subject/الموضوع: AP Chemistry **Unit Title عنوان الوحدة: Unit 1 (Summative) & Unit 6: Thermochemistry** **Grade الصف: 12**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: Unit 1 Summative الحصة 1: التقييم الختامى للوحدة الأولى	• I can apply my understanding of atomic structure, electron configuration, PES and periodic trends to answer AP-style questions independently.	3.4 Support transfer using checklists, organizers, and recurring practice. 4.2 Ensure access to tools and assistive technologies. 7.3 Minimize threats and distractions through predictable structures and varied stimulation.	Unit 1 Review Guide / PPT AP Classroom – Unit 1 Progress Check AP Chemistry Reference Sheet Teams and toddle
Lesson 2: 6.7 Bond Enthalpies الحصة الثانية: طاقة الروابط)إنثالبى الرابطة)	• I can calculate the enthalpy change of a reaction using the average bond enthalpies of bonds broken and bonds formed.	3.3 Guide processing with explicit steps, chunking, and interactive models. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	PPT AP Daily Video – Topic 6.7 (AP Classroom) Bond Enthalpy PRACTICE SHEET Teams and toddle
Lesson 3: 6.8 Hess's Law الحصة الثالثة: قانون هس	• I can use Hess's law to calculate the enthalpy change of a reaction by reversing, scaling and combining a series of steps.	3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 6.2 Support planning with templates, strategy prompts, and modeled thinking. 8.3 Foster collaboration and community with structured group roles and supports.	PPT AP Daily Video – Hess's Law (AP Classroom) Hess's Law PRACTICE SHEET Teams and toddle
Lesson 4: AP Practice Worksheet الحصة الرابعة: ورقة تدريب بأسلوب أسئلة AP	• I can solve AP-style multiple-choice and free-response questions on bond enthalpies and Hess's law, showing clear, justified working.	6.4 Support progress monitoring with rubrics, checklists, and visual progress tools. 8.4 Provide mastery-oriented feedback that emphasizes growth and improvement. 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays.	AP PRACTICE WORKSHEET AP Classroom – Unit 6 Topic Questions AP Chemistry Reference Sheet Teams and toddle

Homework الواجبات	• AP Classroom Topic Questions for 6.7 Bond Enthalpies and 6.8 Hess's Law (due before Session 4). • Finish and self-correct the Session 4 AP-style FRQ using the scoring guideline.	5.3 Build fluency with graduated scaffolds and models. 3.4 Support transfer using checklists, organizers, and recurring practice. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	Toddle
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Session 1 is the Unit 1 Summative test: students need a scientific calculator; the AP reference sheet and periodic table will be provided. Please encourage your child to revise Unit 1 (electron configuration, PES, mass spectra, periodic trends) and to watch the AP Daily videos for Topics 6.7 and 6.8 this week.		

Week # الأسبوع#

Technology Integration دمج التكنولوجيا	AP Classroom (AP Daily videos, Topic Questions, Progress Checks); Kahoot/Quizizz retrieval quiz; digital PPT and practice sheets shared on the school LMS; scientific calculators.
Cross-curricular links الروابط بين المواد الدراسية	Biology – energy in metabolism (breaking and forming bonds in ATP and cellular respiration); Physics – conservation of energy; Mathematics – algebraic manipulation and summation of equations.
External Assessment Integration دمج التقييمات الخارجية	
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Self-management: test preparation, time management under timed AP conditions and managing test stress. Thinking: multistep problem solving. Communication: justifying answers in FRQs. Reflection: self-assessment against AP scoring guidelines.
Standards المعايير	• AP Chemistry CED Unit 1 – Atomic Structure and Properties (Topics 1.1–1.8). • AP Chemistry CED Topic 6.7 – Bond Enthalpies: calculate ΔH of reaction from average bond energies of bonds broken and formed. • AP Chemistry CED Topics 6.8–6.9 – Enthalpy of Formation & Hess's Law: ΔH of a process equals the sum of the ΔH of its steps. • AP Science Practices 5 (Mathematical Routines) and 6 (Argumentation).
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	Inquiry: How can we find the energy change of a reaction that we cannot measure directly in the laboratory? TOK: To what extent can "average" values, such as average bond enthalpies, give us reliable scientific knowledge?
Success Criteria (Measurable) معايير النجاح	The student has met the learning targets by: لقد حقق الطالب/الطالبة أهداف التعلم من خلال: **Session 1 – Unit 1 Summative** • **MUST (يجب):** score at least 60% on the Unit 1 test, correctly writing electron configurations and completing mole / molar-mass calculations. • **SHOULD (ينبغي):** score at least 75%, accurately interpreting mass spectra and photoelectron spectroscopy (PES) data. • **COULD (يمكن):** score 85% or more, fully justifying periodic trends using Coulomb's law and effective nuclear charge in the FRQs. **Session 2 – 6.7 Bond Enthalpies** • **MUST (يجب):** correctly identify bonds broken and formed and calculate ΔH for at least 3 out of 4 simple reactions. • **SHOULD (ينبغي):** calculate ΔH for reactions with multiple, double and triple bonds with the correct sign and units (kJ/mol) in 80% of exit-ticket items. • **COULD (يمكن):** explain why ΔH calculated from average bond enthalpies differs from experimental values. **Session 3 – 6.8 Hess's Law** • **MUST (يجب):** correctly reverse and multiply equations, adjusting the sign and size of ΔH, in at least 3 out of 4 practice steps. • **SHOULD (ينبغي):** combine 2–3 equations to calculate the target ΔH correctly in 80% of practice problems. • **COULD (يمكن):** solve multistep problems using standard enthalpies of formation (ΔH°rxn = ΣΔH°f products – ΣΔH°f reactants) and design their own Hess cycle. **Session 4 – AP Practice Worksheet**

	• **MUST (يجب):** complete the AP-style worksheet, answering at least 60% of the multiple-choice questions correctly. • **SHOULD (ينبغي):** score at least 75% overall, earning most FRQ points with clear calculations, units and significant figures. • **COULD (يمكن):** score 90% or more and peer-mark a classmate's FRQ accurately using the AP scoring guideline.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي History التاريخ
	المجال Area: Conservation & Sustainability – the UAE energy transition
	التطبيق Implementation: Students use bond enthalpies and Hess's law to compare the energy released by burning natural gas (methane) and hydrogen, linking to the UAE's journey from an oil-based economy to the UAE Net Zero 2050 Strategic Initiative and the national hydrogen strategy (e.g., Masdar).